Windows Endpoint Monitoring Lab with Sysmon and Splunk

Executive Summary

This project demonstrates a small SOC monitoring lab built using Sysmon and Splunk. A Windows 10 virtual machine was used to generate system activity, while Splunk was used to collect and analyze the logs.

During the lab, different actions such as command execution, PowerShell usage, and login attempts were performed on the system. These events were then searched and analyzed in Splunk to understand how security analysts investigate system activity using logs.

The goal of the project was to practice basic log monitoring and detection techniques in a simple SOC-style environment.

Lab Monitoring Stack

Victim Endpoint: Windows 10 VM
SIEM: **Splunk Enterprise**

Telemetry: **Sysmon**
Log Collection: **Splunk Universal Forwarder**
Data Source Types Collected:

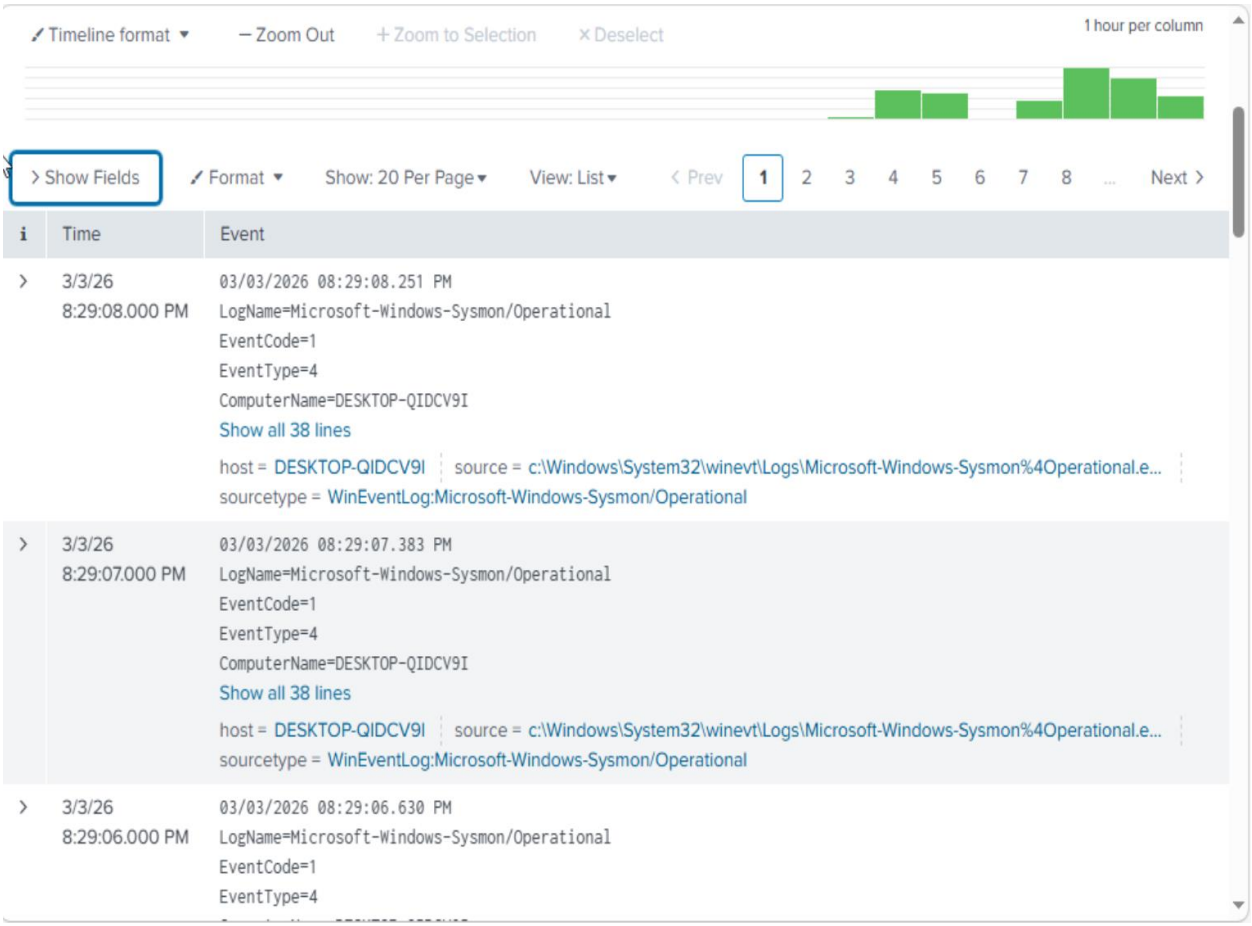
- Sysmon Operational logs
- Windows Security authentication logs
- Windows System/Application logs

Phase 1 — Telemetry Verification (Start of Project)

Goal: Confirm endpoint telemetry is being collected correctly.

Query used:

```
index=main EventCode=1
```



Sysmon process creation telemetry successfully ingested into Splunk SIEM.

Phase 2 — Baseline Process Activity

Goal: Establishing what “normal” looks like before attack simulation.

Query used:

```
index=main EventCode=1
| stats count by Image
| sort – count
```

Events Patterns Statistics (110) Visualization

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Image	count
C:\Program Files\Splunk\bin\splunk-optimize.exe	11508
C:\Program Files\Splunk\bin\python3.13.exe	550
C:\Program Files\Splunk\bin\splunk-powershell.exe	480
C:\Program Files\Splunk\bin\splunkd.exe	476
C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	342
C:\Windows\System32\cmd.exe	290
C:\Program Files\Splunk\bin\splunk-MonitorNoHandle.exe	240
C:\Program Files\Splunk\bin\splunk-admon.exe	240
C:\Program Files\Splunk\bin\splunk-netmon.exe	240
C:\Program Files\Splunk\bin\splunk-regmon.exe	240
C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe	171
C:\Program Files\SplunkUniversalForwarder\bin\splunk-winevtlog.exe	171
C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe	170
C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe	170

Baseline process frequency captured to support anomaly detection.

Phase 3 — Suspicious Command Execution Simulation

Goal: Simulate command-line reconnaissance activity commonly performed by attackers after gaining initial access to a system.

To generate realistic telemetry command-line commands such as **whoami**, **ipconfig**, were executed on the Windows 10 endpoint.

Action (Windows 10 VM):

Query used:
index=main EventCode=1
| search CommandLine="*whoami*" OR CommandLine="*ipconfig*"
| table _time Image CommandLine ParentImage

✓ **10 events** (before 3/4/26 1:11:30.000 PM) No Event Sampling ▾ Job ▾ Fast Mode ▾

Events Patterns **Statistics (10)** Visualization

Show: 10 Per Page ▾ Format ▾ Preview: On

_time	ComputerName	Image	CommandLine	ParentImage
2026-03-04 13:07:46	DESKTOP-QIDCV9I	C:\Windows\System32\ipconfig.exe	ipconfig	C:\Windows\System32\cmd.exe
2026-03-04 13:07:40	DESKTOP-QIDCV9I	C:\Windows\System32\whoami.exe	whoami	C:\Windows\System32\cmd.exe
2026-03-03 20:54:11	DESKTOP-QIDCV9I	C:\Windows\System32\ipconfig.exe	ipconfig	C:\Windows\System32\cmd.exe
2026-03-03 20:54:06	DESKTOP-QIDCV9I	C:\Windows\System32\whoami.exe	whoami	C:\Windows\System32\cmd.exe
2026-03-03 20:48:48	DESKTOP-QIDCV9I	C:\Windows\System32\ipconfig.exe	ipconfig	-
2026-03-03 20:48:40	DESKTOP-QIDCV9I	C:\Windows\System32\whoami.exe	whoami	-
2026-03-03 21:32:14	DESKTOP-QIDCV9I	C:\Windows\System32\ipconfig.exe	ipconfig	C:\Windows\System32\cmd.exe
2026-03-03 21:32:08	DESKTOP-QIDCV9I	C:\Windows\System32\whoami.exe	whoami	C:\Windows\System32\cmd.exe
2026-03-03 21:06:18	DESKTOP-QIDCV9I	C:\Windows\System32\ipconfig.exe	ipconfig	C:\Windows\System32\cmd.exe
2026-03-03 21:06:06	DESKTOP-QIDCV9I	C:\Windows\System32\whoami.exe	whoami	C:\Windows\System32\cmd.exe

This query filters process creation events to show only the commands that were executed during the test.The search results show executions of command-line reconnaissance commands including **whoami** and **ipconfig**, which were successfully captured by Sysmon and indexed into Splunk.

Phase 4 — PowerShell Activity Simulation

Goal: The goal of this phase was to generate PowerShell activity on the endpoint and detect the commands using Splunk search queries.

Query used:

```
index=main EventCode=1
| search CommandLine="*Get-Process*" OR CommandLine="*Get-Service*" OR
CommandLine="*systeminfo*"
| table _time ComputerName Image CommandLine ParentImage
```

✓ **4 events** (before 3/4/26 1:49:52.000 PM) No Event Sampling ▾ Job ▾ Fast Mode ▾

Events Patterns **Statistics (4)** Visualization

Show: 10 Per Page ▾ Format ▾ Preview: On

_time	ComputerName	Image	CommandLine	ParentImage
2026-03-04 13:44:10	DESKTOP-QIDCV9I	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell -Command "systeminfo"	-
2026-03-04 13:40:08	DESKTOP-QIDCV9I	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell -Command "Get-Service"	-
2026-03-04 13:39:54	DESKTOP-QIDCV9I	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell -Command"Get-Service"	-
2026-03-04 13:39:34	DESKTOP-QIDCV9I	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell -Command "Get-Process	-

This query filters process creation events to show only the commands that were executed during the test. The results confirm that PowerShell activity was successfully captured and can be investigated using Splunk.

Phase 5 — Authentication Monitoring (Credential Abuse Signals)

Goal: The goal of this phase was to monitor login activity and detect successful and failed authentication attempts on the Windows system.

Successful logins:

Query used:

```
index=main EventCode=4624
```

✓ **937 events** (before 3/4/26 2:27:24.000 PM) No Event Sampling ▾ Job ▾ Fast Mode ▾

Events Patterns **Statistics (937)** Visualization

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_time	ComputerImage	EventCode	Account_Name	Logon_Type	Source_Network_Address
2026-03-04 14:26:02.000		4624	DESKTOP-QIDCV9I$ SYSTEM	5	-
2026-03-04 14:26:02.429		4624	DESKTOP-QIDCV9I$ SYSTEM	5	-
2026-03-04 14:24:17.000		4624	DESKTOP-QIDCV9I$ waveesports95@gmail.com	7	-
2026-03-04 14:24:17.000		4624	DESKTOP-QIDCV9I$ waveesports95@gmail.com	7	-
2026-03-04 14:23:48.000		4624	DESKTOP-QIDCV9I$ SYSTEM	5	-
2026-03-04 14:23:42.000		4624	DESKTOP-QIDCV9I$ SYSTEM	5	-
2026-03-04 14:24:17.478		4624	DESKTOP-QIDCV9I$ waveesports95@gmail.com	7	-
2026-03-04 14:24:17.478		4624	DESKTOP-QIDCV9I$ waveesports95@gmail.com	7	-
2026-03-04		4624	DESKTOP-QIDCV9I$	5	-

Successful authentication events reviewed for abnormal patterns.

Failed logins:

Query used:

```
index=main EventCode=4625
```

✓ **14 events** (before 3/4/26 2:29:24.000 PM) No Event Sampling ▾ Job ▾ Fast Mode ▾

Events Patterns **Statistics (14)** Visualization

Show: 10 Per Page ▾ Format ▾ Preview: On < Prev 1 2 Next >

_time	ComputerImage	EventCode	Account_Name	Logon_Type	Source_Network_Address
2026-03-04 14:24:03.000		4625	DESKTOP-QIDCV9I$ -	2	127.0.0.1
2026-03-04 14:24:01.000		4625	DESKTOP-QIDCV9I$ -	2	127.0.0.1
2026-03-04 14:23:59.000		4625	DESKTOP-QIDCV9I$ -	2	127.0.0.1
2026-03-04 14:23:57.000		4625	DESKTOP-QIDCV9I$ -	2	127.0.0.1
2026-03-04 14:24:03.467		4625	DESKTOP-QIDCV9I$ -	2	127.0.0.1
2026-03-04 14:24:01.702		4625	DESKTOP-QIDCV9I$ -	2	127.0.0.1
2026-03-04 14:23:59.959		4625	DESKTOP-QIDCV9I$ -	2	127.0.0.1
2026-03-04 14:23:57.462		4625	DESKTOP-QIDCV9I$ -	2	127.0.0.1
2026-03-04		4625			

Failed authentication attempts monitored as potential unauthorized access activity.

Phase 6 — Detection Engineering (Simple Rules)

Goal: Create practical SOC detections from observed behavior.

6.1 Detect repeated PowerShell usage

Query used:

```
index=main EventCode=1 Image="*powershell*"
| stats count by ComputerName, User, CommandLine
| sort - count
```

✓ 28,942 events (before 3/4/26 2:41:01.000 PM) No Event Sampling ▾ Job ▾ Fast Mode ▾

Events Patterns Statistics (182) Visualization

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Image	count
C:\Program Files\Splunk\bin\splunk-optimize.exe	19590
C:\Program Files\Splunk\bin\python3.13.exe	902
C:\Program Files\Splunk\bin\splunk-powershell.exe	794
C:\Program Files\Splunk\bin\splunkd.exe	702
C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	658
C:\Windows\System32\cmd.exe	404
C:\Program Files\Splunk\bin\splunk-MonitorNoHandle.exe	398
C:\Program Files\Splunk\bin\splunk-admon.exe	397
C:\Program Files\Splunk\bin\splunk-netmon.exe	397
C:\Program Files\Splunk\bin\splunk-regmon.exe	397

This query shows which processes are executed most frequently on the system. Monitoring process activity helps analysts identify unusual programs or suspicious command execution.

6.2 Detect suspicious command shells

Query used:

```
index=main EventCode=1 (Image="*cmd.exe" OR Image="*powershell.exe")
| table _time User Image CommandLine ParentImage
```

✓ 27 events (before 3/4/26 5:05:19.000 PM) No Event Sampling ▾ Job ▾ Smart Mode ▾

Events Patterns Statistics (27) Visualization

Show: 20 Per Page ▾ Format ▾ Preview: On < Prev 1 2 Next >

_time	ComputerName	User	Image	CommandLine
2026-03-04 17:03:01	DESKTOP-QIDCV9I	NOT_TRANSLATED DESKTOP-QIDCV9I\wavee	C:\Windows\System32\HOSTNAME.EXE	hostname
2026-03-04 17:02:58	DESKTOP-QIDCV9I	NOT_TRANSLATED DESKTOP-QIDCV9I\wavee	C:\Windows\System32\tasklist.exe	tasklist
2026-03-04 17:02:46	DESKTOP-QIDCV9I	NOT_TRANSLATED DESKTOP-QIDCV9I\wavee	C:\Windows\System32\NETSTAT.EXE	netstat -ano
2026-03-04 17:01:33	DESKTOP-QIDCV9I	NOT_TRANSLATED DESKTOP-QIDCV9I\wavee	C:\Windows\System32\systeminfo.exe	systeminfo
2026-03-04 17:01:18	DESKTOP-QIDCV9I	NOT_TRANSLATED DESKTOP-QIDCV9I\wavee	C:\Windows\System32\ipconfig.exe	ipconfig
2026-03-04 17:00:46	DESKTOP-QIDCV9I	NOT_TRANSLATED DESKTOP-QIDCV9I\wavee	C:\Windows\System32\whoami.exe	whoami

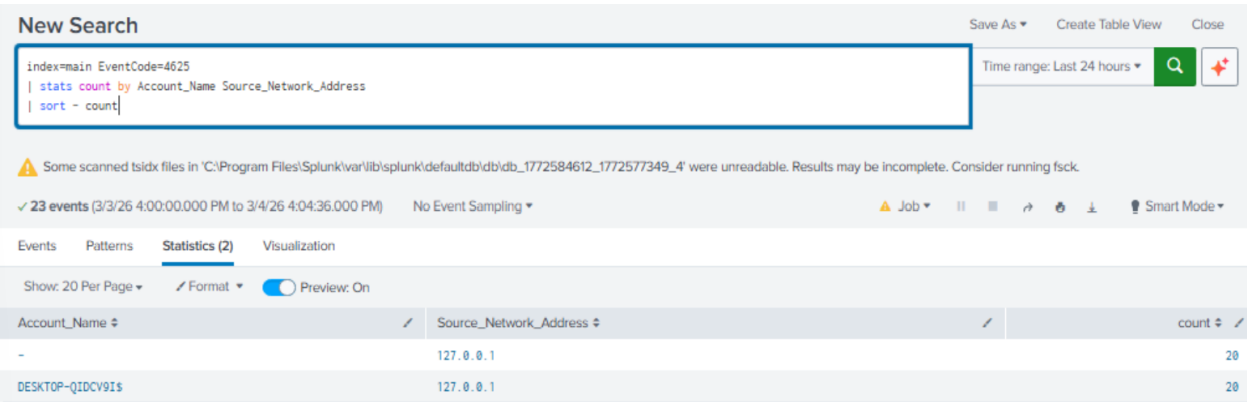
This query filters logs to display command shell and PowerShell execution events. Since command shell and PowerShell is commonly used by attackers to run scripts and gather system information, monitoring PowerShell activity is important for detecting suspicious behavior.

6.3 Failed Login Attempts For Each Account

Query used:

Index=main EventCode=4625
| stats count by Account_Name Source_Address

| sort - count



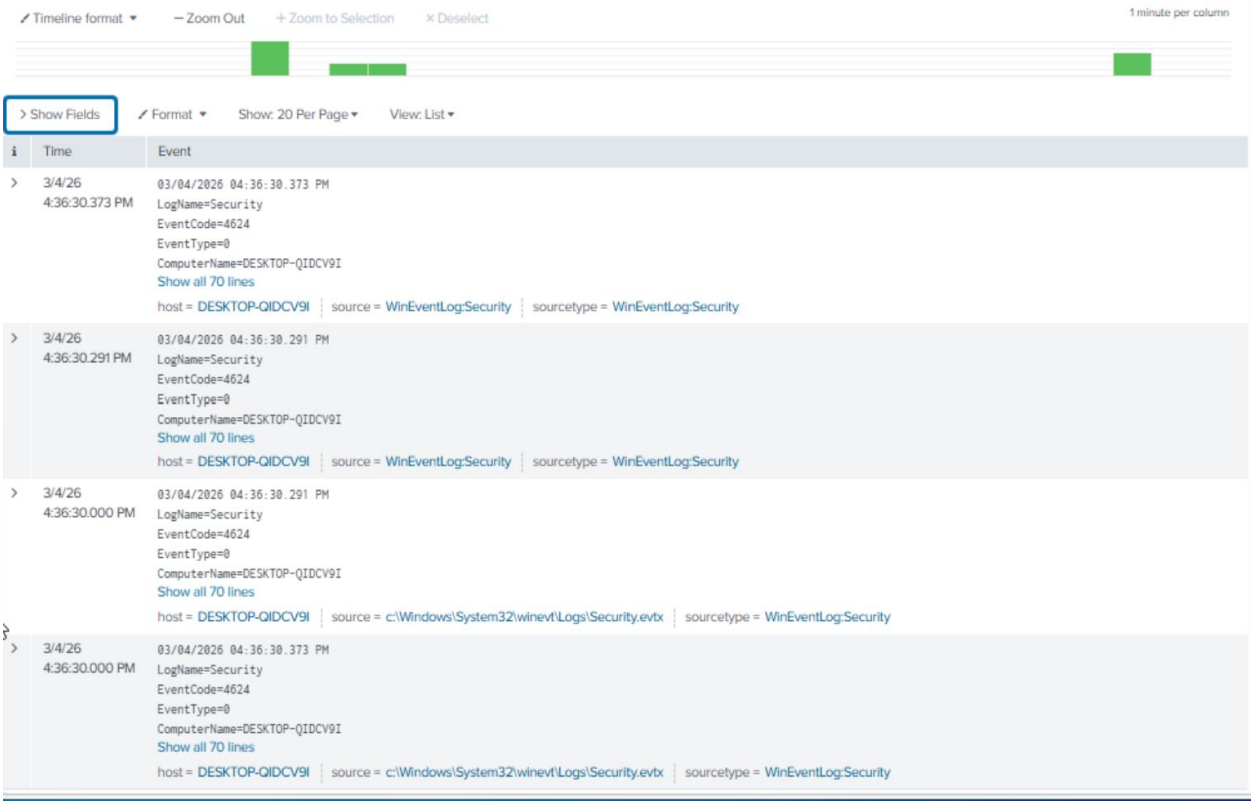
This query counts the number of failed login attempts for each account. Multiple failed attempts from the same user or system may indicate a brute-force login attempt or unauthorized access activity.

Phase 7 — Incident Timeline Reconstruction

Goal: The goal of this phase was to review the timeline of system activity during the investigation.

Query used:

```
index=main (EventCode=1 OR EventCode=4624 OR EventCode=4625)
| sort _time
```



The search results show authentication events recorded in Windows Security logs. Event ID 4624 indicates successful logins to the system. Reviewing these events helps analysts understand when users accessed the system and correlate authentication activity with other events.

Phase 8 — Detection Alert Simulation

Goal:

Create a simple SIEM alert to detect suspicious command execution.

Query used:

```
index=main EventCode=1
| search CommandLine="*whoami*" OR CommandLine="*ipconfig*" OR
CommandLine="*net user*" OR CommandLine="*systeminfo*" OR
```

```
CommandLine="*netstat*" OR CommandLine="*tasklist*"
| stats count by ComputerName User CommandLine
```

✓ 19 events (before 3/4/26 5:55:24.000 PM) No Event Sampling ▾ Job ▾ Smart Mode ▾

Events Patterns Statistics (16) Visualization

Show: 20 Per Page ▾ Format ▾ Preview: On

ComputerName	User	CommandLine	count
DESKTOP-QIDCV9I	DESKTOP-QIDCV9I\wavee	"C:\Windows\system32\NETSTAT.EXE" -ano	1
DESKTOP-QIDCV9I	DESKTOP-QIDCV9I\wavee	"C:\Windows\system32\systeminfo.exe"	2
DESKTOP-QIDCV9I	DESKTOP-QIDCV9I\wavee	ipconfig	6
DESKTOP-QIDCV9I	DESKTOP-QIDCV9I\wavee	netstat -ano	1
DESKTOP-QIDCV9I	DESKTOP-QIDCV9I\wavee	powershell -Command "systeminfo"	1
DESKTOP-QIDCV9I	DESKTOP-QIDCV9I\wavee	systeminfo	1
DESKTOP-QIDCV9I	DESKTOP-QIDCV9I\wavee	tasklist	1
DESKTOP-QIDCV9I	DESKTOP-QIDCV9I\wavee	whoami	6
DESKTOP-QIDCV9I	NOT_TRANSLATED	"C:\Windows\system32\NETSTAT.EXE" -ano	1
DESKTOP-QIDCV9I	NOT_TRANSLATED	"C:\Windows\system32\systeminfo.exe"	2
DESKTOP-QIDCV9I	NOT_TRANSLATED	ipconfig	6
DESKTOP-QIDCV9I	NOT_TRANSLATED	netstat -ano	1
DESKTOP-QIDCV9I	NOT_TRANSLATED	powershell -Command "systeminfo"	1
DESKTOP-QIDCV9I	NOT_TRANSLATED	systeminfo	1
DESKTOP-QIDCV9I	NOT_TRANSLATED	tasklist	1
DESKTOP-QIDCV9I	NOT_TRANSLATED	whoami	6

This detection query identifies reconnaissance commands commonly executed by attackers after gaining access to a system. Monitoring these commands helps security analysts detect early signs of system enumeration activity.

Conclusion

In this project, a small SOC monitoring lab was created using Windows, Sysmon, and Splunk. System logs were collected and analyzed to monitor command execution, PowerShell activity, and authentication events.

During the investigation, different types of activity such as command-line execution, PowerShell usage, and failed login attempts were successfully detected using Splunk search queries.

This project demonstrates how security analysts can use log monitoring and detection queries to identify suspicious behavior and investigate potential security incidents.

THANK YOU!